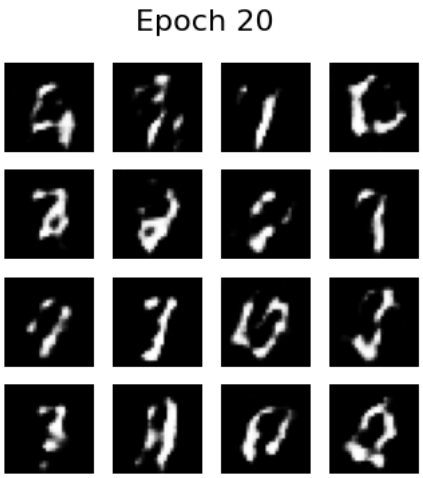
```
--- Part 1: Loading and Preprocessing the MNIST Dataset ---
Normalized training data shape: (60000, 28, 28, 1)
Example of a normalized pixel value: -1.0

--- Part 2: Building the GAN Components ---

--- Generator Model Summary ---

--- Discriminator Model Summary ---

--- Beginning GAN Training ---
Epoch 1/20 - Generator Loss: 0.6768, Discriminator Loss: 1.1637
Epoch 2/20 - Generator Loss: 0.7540, Discriminator Loss: 1.3254
Epoch 3/20 - Generator Loss: 0.8536, Discriminator Loss: 1.2518
Epoch 4/20 - Generator Loss: 0.8303, Discriminator Loss: 1.2697
Epoch 5/20 - Generator Loss: 0.7852, Discriminator Loss: 1.3430
Epoch 6/20 - Generator Loss: 0.8433, Discriminator Loss: 1.2608
Epoch 7/20 - Generator Loss: 0.8500, Discriminator Loss: 1.2787
Epoch 8/20 - Generator Loss: 0.8037, Discriminator Loss: 1.3143
Epoch 9/20 - Generator Loss: 0.8954, Discriminator Loss: 1.2486
Epoch 10/20 - Generator Loss: 0.9607, Discriminator Loss: 1.1917
Epoch 11/20 - Generator Loss: 0.9710, Discriminator Loss: 1.1546
Epoch 12/20 - Generator Loss: 1.0153, Discriminator Loss: 1.1516
Epoch 13/20 - Generator Loss: 1.0767, Discriminator Loss: 1.1456
Epoch 14/20 - Generator Loss: 1.0831, Discriminator Loss: 1.1132
Epoch 15/20 - Generator Loss: 1.0662, Discriminator Loss: 1.1257
Epoch 16/20 - Generator Loss: 1.1792, Discriminator Loss: 1.0287
Epoch 17/20 - Generator Loss: 1.1277, Discriminator Loss: 1.1004
Epoch 18/20 - Generator Loss: 1.1097, Discriminator Loss: 1.0886
Epoch 19/20 - Generator Loss: 1.0837, Discriminator Loss: 1.1078
Epoch 20/20 - Generator Loss: 1.1708, Discriminator Loss: 1.0407
```




```
--- Training complete. Generating final images. ---
```

